

Technical Research Report

Casey Stijlaart

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1. Document history

Table 1. Document History

Version	Date	Status	Author	Description	Remarks
0.1	03-09-2026	Draft Version	Casey Stijlaart	Draft Version	Initial document layout, non-camera sensing techniques chapter
0.2	04-09-2026	Draft Version	Casey Stijlaart	Draft Version	Finalisation of non-camera sensing techniques

2. Non-Camera Person Position & Height Sensing Techniques

2.1. Introduction

This chapter addresses the first technical research question defined in the project plan (see **Research Questions**): "Which combination of non-camera sensors provides sufficient 3D spatial resolution to distinguish adult and child height categories within a room-scale interactive space?"

Per the project assignment (see **Assignment**), the target system estimates a single person's real-time 3D position and height within the Holobox interaction space without cameras, and exposes that estimate to the Holobox's Unreal Engine pipeline so the Virtual Human can adapt its behaviour accordingly. This chapter's scope is the question that decision depends on; which non-camera sensing technologies exist, what they have actually been shown to do in the literature, and which module aligns the most with the project's requirements and possible future extensibility.

In scope for this chapter various aspects will be taken into consideration and discussed. Such aspects involve a literature-backed study of candidate non-camera sensing technologies, the criteria a viable approach has to satisfy, a comparison of specific sensors and sensor combinations against those criteria using current hardware/cost data, and a recommended sensor choice with a staged plan for de-risking it, detailed in the sections that follow.

What is not included in the scope for this chapter is the specifics of embedded implementation and low-latency signal-processing pipeline design, training and deploying the on-device ML height-classification model, and the structure of the Unreal Engine output interface are addressed by other technical sub-questions in the project plan's research questions. This chapter recommends which sensor(s) to use and how to validate that choice, not how the resulting embedded software is built. Per **Scope and Preconditions**, a fully polished prototype ready for a non-controlled environment is explicitly out of scope for this project, since it is exploratory by nature; the recommendation in this chapter reflects that by ending in a staged prototyping plan rather than a final hardware order.

Two aspects of the research question are also narrowed for this stage. Distinguishing adult and child height **categories**, as literally asked above, is treated as a possible later refinement rather than a hard requirement now: a general height estimate is sufficient for this stage (see Candidate Sensor Technologies). Multi-person tracking is explicitly a future-facing requirement in the project assignment, not part of this project's build scope; it is addressed here only as one of the criteria technologies are evaluated against (see Evaluation Criteria), not as a capability this stage implements. If time does allow it, these aspects will be looked further into and possibly integrated (to some extent).

2.2. Research Approach

This chapter follows the Library and Lab strategies of the DOT-framework identified for this research question in **Research Methods**: available product analysis, literature study, document analysis, and bench experimentation.

2.2.1. Library: Literature Study

Candidate Sensor Technologies gathers and verifies peer-reviewed literature for each non-camera sensing family against RQ1's position/height requirement and, separately, against the multi-person active-speaker problem relevant to RQ4. The literature is gathered in two separate passes because the two problems draw on largely different bodies of research: RQ1 sits in the radar/thermal/RF position-and-pose-estimation literature, while RQ4's active-speaker problem sits closer to acoustic and vibrometry-based speaker localisation, so a single combined search would have missed sources specific to either.

2.2.2. Lab: Available Product Analysis

Alongside the literature, the same chapter narrows the literature-backed technology families down to specific, currently sourceable hardware modules, checked against live distributor listings (Digi-Key, Mouser, Adafruit, and others) rather than assumed pricing. An unverified cost estimate is not a usable basis for a hardware decision, nor for the budget-approval process in **Finance and Risks**. This step matters because literature backing and product availability are two different questions. A sensing technique can be well-published without there being a specific, currently purchasable module that implements it at a reasonable price, or at all.

2.2.3. Lab: Bench Experimentation Strategy

The recommended path into bench experimentation commits to mmWave radar directly, and manages implementation risk by splitting the build into capability-based phases rather than assuming the full pipeline works at once: position tracking first, then height estimation, then multi-person clustering, then speaker detection, each building on an already-working previous phase. This choice of experimentation strategy is deliberate, for four reasons:

- **Incremental validation.** Each phase adds exactly one new capability on top of a phase that already works end to end, so a failure is isolated to whichever capability is currently being added, rather than surfacing somewhere in an untested, fully assembled system.
- **No wasted integration effort.** Because the sensor is the same mmWave hardware throughout, the UART bring-up, point-cloud parsing, and Unreal Engine bridge built in the first phase are reused, not redone, in every phase after it.
- **Follows the research question sequence.** Position and height tracking (RQ1) are validated before multi-person clustering and speaker detection (RQ4) are added, so bench experimentation tracks the same sequence the research questions themselves are structured in, rather than an arbitrary build order.
- **Reflects feature priority.** Not every phase is equally critical: position and height tracking

are core RQ1 requirements, while multi-person clustering and speaker detection are RQ4 extensions that improve the system without being required for the current single-person scope, so the build order also matches what is essential versus what is a later improvement.

A working fallback still exists at each stage: if a later phase's capability underperforms, the earlier phase's version remains a functioning system rather than a dead end, which mirrors the fallback the project's own risk register anticipates for both sensor selection and on-device ML performance.

2.3. Evaluation Criteria

Before diving into available sensor modules, it is of great importance to understand the requirements they need to meet. Thus the comparison can be set out against explicit, justified requirements rather than an implicit sense of what "good" looks like. Each criterion below is traced to either a technical sub-question in **Research Questions**, a key feature named in the **Assignment**, or a project constraint from **Finance and Risks**, rather than to a generic notion of sensor quality.

Table 2. Evaluation Criteria

Criterion	What it means here	Why it matters
Height-estimation capability	The sensor (alone or combined with a second sensor) must produce an actual height reading, not just an in-plane (x, y) position.	Follows from RQ1, which asks for "sufficient 3D spatial resolution... to distinguish adult and child height categories." Per the project's current scope, however, a general height estimate is treated as sufficient for now, with category classification as a possible later refinement. A technology with no published height precedent at all is a bigger open research question than one with a direct precedent.
Position accuracy and coverage	Position error and effective range must be usable at the Holobox's actual room-scale dimensions, not just in a research lab's test setup.	RQ1's spatial-resolution requirement extends to (x, y) tracking as well as height; a sensor that cannot reliably cover the interaction space is not usable regardless of its height capability.

Criterion	What it means here	Why it matters
Real-time embedded feasibility	Must run at interactive latency on an embedded platform, not require offloading to an external server. Total pipeline latency (sensor plus processing plus output) is targeted at under roughly 100 ms to feel instantaneous, and must not exceed roughly 1 second, the point past which a delay becomes noticeable and the interaction starts to feel unnatural.	RQ2 requires "robust, low-latency position estimates" on an embedded platform, and the user/interaction sub-questions ask directly what latency the Virtual Human's reaction needs to feel "natural and immediate, rather than delayed or robotic." The 100 ms / 1 s thresholds follow Nielsen's (1993) classic human-computer interaction response-time limits, building on Miller (1968), used here as a working target since this project's own latency sub-question has not yet been answered empirically.
Sensor combination / fusion compatibility	Whether the sensor can realistically be paired with a second sensor to jointly produce position and height, without a fundamentally incompatible operating principle or output format.	Named directly as a key feature in the project assignment: "ML-based or rule-based sensor combination of two or more non-camera sensors... to achieve robust position and height estimation." A technology that cannot realistically be combined with a second sensor forecloses that option later, even if it performs well alone.
Multi-person extensibility	Should not fundamentally break, or require a different sensing approach altogether, if more than one person enters the space later.	RQ4 requires the architecture to remain "extensible to multi-person scenarios," and the Assignment separately names multi-person tracking as "a future-facing requirement," even though single-person tracking is the current scope.

Criterion	What it means here	Why it matters
Data minimisation / privacy	Output should not be capable of identifying a specific individual, and should capture the minimum data needed for position and height.	Directly required by the ethical/legal sub-questions: primary visitors are children aged 8-12, and the system must avoid capturing or storing personally identifiable information under AVG/GDPR.
Literature and community backing	Prior published work (ideally peer-reviewed) should exist showing the specific capability required, not just the technology in general.	Reduces the risk of committing project time to proving something has never been demonstrated at all, as distinct from the normal risk of not knowing whether it will work well enough .
Hardware cost	Cost of the sensor and any second sensor it must be paired with, checked against current, sourceable listings.	No project budget has yet been agreed with MindLabs, so cost is a real constraint on what can be selected, not a tie-breaker between otherwise-equal options.
Power consumption and footprint	Physical size and power draw must fit within the Holobox's existing hardware, without requiring new infrastructure.	The Holobox is an existing installation; the sensing system is being added to it, not designed around a blank slate.

NOTE

Sensor output rate is only one component of total pipeline latency, since processing, sensor fusion, and the Unreal Engine transmission step all add more on top, but it sets a floor worth checking early.

These criteria are not weighted equally, and are not applied as a hard pass/fail filter. Height-estimation capability and data minimisation are treated as closer to hard requirements, since they follow directly from RQ1 and the ethical/legal sub-questions; the remaining criteria (sensor combination, cost, power/footprint) are weighed against each other in the chapter that follows, since a technology that is merely more expensive or harder to pair with a second sensor is not disqualified the way one with no height precedent or an identifying output effectively is.

2.4. Candidate Sensor Technologies

This chapter surveys non-camera sensing technologies capable of device-free person detection, localisation, and (where the literature supports it) height estimation, evaluated against the technical sub-questions in [Project Plan - Research Questions](#), primarily the requirement for sufficient 3D spatial resolution to estimate a person's height, embedded-platform feasibility, and the data-minimisation/privacy constraints that follow from the primary visitors being children aged 8-12. Distinguishing adult and child height is not a hard requirement at this stage. A general height estimate is sufficient for now, with category classification treated as a possible refinement once a candidate technology is selected.

Each technology below is covered in two parts: what the literature has actually shown it can do, and, for the technically viable candidates, how that holds up once weighed against the criteria set out earlier, specific hardware modules, current cost, and pros and cons. The chapter closes with a comparison summary, a criteria matrix across every approach discussed, and the resulting recommendation.

Ngamakeur et al. (2020) survey device-free indoor localisation and tracking, framing this same comparison at a broader scale. It structures device-free sensing into a pipeline ([Figure 1](#)) that compares Wi-Fi, radar, infrared, and thermal approaches against each stage. That pipeline view is used below to organise what each candidate technology has actually been shown to do in the literature, rather than what it could in principle do.



Figure 1. Device-free Pipeline

2.4.1. Individual Position & Height Localisation

This section covers single-person, device-free position and height sensing, the core of RQ1.

Time-of-Flight & LiDAR

2D LiDAR measures range by timing a laser pulse's reflection while scanning across a single horizontal plane, producing an (x, y) point cloud at the height the unit is mounted. This makes it well suited to in-plane position and leg/torso-level tracking, but a single fixed-height scan plane does not by itself yield a height measurement. Retrieving the height of an individual would require either multiple units mounted at different heights, an angled/tilted scan, or a full 3D ToF/LiDAR unit instead.

Within its native (x, y) tracking role, 2D LiDAR performs well: Bouazizi et al. (2024) fuse multiple 2D LiDAR units with a ConvLSTM model for indoor activity detection and show the approach tolerates furniture occlusion, a realistic condition for a room-scale interactive space. At the single-scan-plane level, Enoki et al. (2024) show waist-height contour data is sufficient for person identification and activity estimation using DBSCAN clustering with a deep learning classifier, which indicates the sensing modality carries enough per-person shape information

for classification tasks even from one plane. For the extensibility sub-question (RQ4), Alam et al. (2021) document, in PALMAR's related-work analysis, "crossover ambiguity" as the recurring failure mode of 2D LiDAR when multiple people's tracks pass close together, which is the concrete risk to plan around if this technology is carried forward into a multi-person scenario later.

On latency, the representative module identified below, the RPLIDAR A2M8, has a manufacturer-documented scan rate of 5-15 Hz (typically 10 Hz, i.e. 600 rpm), independently adjustable from its sample rate of up to roughly 4000 samples per second per Slamtec's datasheet. This scan rate, not the faster per-sample rate, is what limits how often a full (x, y) frame around the sensor is available.

Implication for RQ1: 2D LiDAR alone does not produce a height estimate. It would need to be paired with a second height-sensing modality or reconfigured (multi-height mounting, tilt-scanning, or a 3D ToF unit) to contribute a height signal, which should be weighed against its strong in-plane positional accuracy. This chapter carries that pairing forward as part of **2D LiDAR + Thermal Fusion**, below.

Pros:

- Strong in-plane position accuracy and furniture-occlusion tolerance when multiple units are fused (Bouazizi et al.).
- A single scan plane already carries enough per-person shape information for identification and activity classification (Enoki et al.), not just raw (x, y) position.
- High angular sampling rate (up to ~4000 samples/sec per Slamtec's datasheet), independent of the slower scan rate that paces full-frame updates.
- Mature, widely available consumer/robotics hardware (RPLIDAR family) with broad community and driver support.

Cons:

- Produces no height reading from a single fixed-height scan plane; needs multi-height mounting, a tilted scan, or a full 3D ToF/LiDAR unit to contribute a height signal at all.
- "Crossover ambiguity" is a documented failure mode when multiple people's tracks pass close together (Alam et al., PALMAR related work), a concrete risk for RQ4 multi-person extensibility.
- Scan rate of 5-15 Hz (typically 10 Hz) is on the slower end of the candidates reviewed, and sets the update-frequency ceiling regardless of the faster per-sample rate.
- Carrying height forward at all means accepting a second sensor (thermal, see **2D LiDAR + Thermal Fusion**) or added mechanical complexity (multi-height mounting/tilt-scanning), not a same-sensor fix.

mmWave Radar

mmWave radar has an extensive body of research literature covering both position tracking

and skeletal/joint-level pose estimation. Point-cloud outputs from consumer mmWave radar are also inherently non-identifying (no recognisable imagery), which aligns with the ethical/legal sub-questions on minimising personally identifiable data collected from child visitors.

For position tracking, Huang et al. (2021) combine clutter removal, clustering, and Kalman filtering for multi-person tracking running in real time on a Raspberry Pi 4. Pegoraro and Rossi (2022) replace the Kalman filter with a CNN-RNN tracker with uncertainty estimation, reporting position error down to roughly 7.6 cm.

More directly relevant to the height-estimation requirement is the body of work on mmWave-based pose estimation: Zhu et al.'s ProbRadarM3F (2024) recovers 14 body keypoints from radar point clouds via probability-map-guided multi-format feature fusion, and Zheng et al. (2025) survey related keypoint-estimation work (mmPose, mm-Pose, RadHAR) in their related-work section, giving a further citation trail if deeper comparison is needed. Supporting datasets exist for this line of work: Su et al.'s mmFree-Pose dataset (2024) provides a camera-free radar dataset with 23-joint ground truth captured via motion capture, and Rahman et al.'s MMVR (2024) is a multi-view radar benchmark for indoor perception that references HuPR and other radar pose-estimation benchmarks. As a keypoint output includes head/shoulder joints, this is the most direct existing path from raw sensor data to an actual height estimate, rather than a proxy such as blob size.

On the positioning-fundamentals side, Almalioglu et al.'s Milli-RIO (2019) demonstrates centimetre-level ego-motion/indoor localisation using a single low-cost mmWave chip plus an IMU, which is a useful data point for embedded cost and compute feasibility (RQ2) independent of the pose-estimation question.

On latency specifically, the two position-tracking papers above report frame rates on the radar hardware they each actually used, rather than the TI IWR6843 recommended below: Pegoraro and Rossi (2022) report their INRAS RadarLog-based tracker outputting radar-Doppler maps at 15 fps (roughly 67 ms per frame), and Huang et al. (2021) target a 50 ms (20 fps) frame budget on their IWR1642BOOST-based pipeline specifically so the Raspberry Pi 4 does not fall behind and drop frames. For the IWR6843 chip itself, TI's datasheet supports frame rates up to roughly 76 Hz, but TI's own demo/application configurations typically run at 10-20 Hz (50-100 ms per frame); see the latency figures already established in Evaluation Criteria. These are three different radar chips reporting broadly similar tens-of-milliseconds frame periods, which is a reasonable indication of what mmWave point-cloud tracking can achieve in general, but not a benchmark of the IWR6843 specifically.

Implication for RQ1: mmWave radar's literature includes published keypoint/pose-level output alongside demonstrated real-time embedded operation, and its device-free, non-identifying output profile fits the privacy sub-questions well.

Available product analysis narrows this to a 60 GHz point-cloud radar module built around the TI IWR6843 chip, from the same TI mmWave SDK/point-cloud family as the position-tracking literature above even though the exact chip cited differs paper to paper, over two alternatives considered and set aside: low-cost 24 GHz presence sensors (e.g. the HLK-LD2410 class of module), which output a binary presence flag rather than a point cloud and cannot support

position or height estimation at all; and undocumented generic radar modules, which lack the reference software and community support needed to reach a working pipeline within the project timeline. Several evaluation-board variants exist within the IWR6843 family itself, at different price points and field-of-view characteristics; **mmWave Module Selection** compares them directly and picks one to carry forward, so this chapter treats "TI IWR6843" as the candidate and defers the specific-variant decision to that chapter.

Pros:

- Strongest literature base of any candidate: real-time multi-person tracking (Huang et al.), keypoint-level pose/height estimation (Zhu et al., Zheng et al.).
- Demonstrated real-time operation on a Raspberry Pi 4.
- Point-cloud output is inherently non-identifying, fitting the project's data-minimisation requirements.
- Extends to the RQ4 active-speaker problem via the same sensor family (mmWave vibrometry), without a second dedicated device.
- Point clouds naturally separate multiple people, supporting future multi-person extensibility.

Cons:

- By far the most expensive candidate (€160-185 for the IWR6843AOPEVM variant compared in **mmWave Module Selection**, roughly three to four times the thermal array's sensor cost).
- Height estimation still requires extra work: a choice between a blob-size heuristic or a keypoint-extraction model, rather than a single established method.
- No project budget has yet been agreed with MindLabs, so this cost is a real, unresolved risk rather than a formality.

Wi-Fi CSI Sensing

Wi-Fi Channel State Information (CSI) sensing infers presence and movement from fluctuations in the wireless channel caused by a body moving through it. It requires no dedicated sensor beyond commodity Wi-Fi hardware, which is attractive for cost, but the gathered literature is consistent that it trades away spatial resolution and stability. CSI is sensitive to environmental changes (furniture, multipath, other RF sources), and reported positioning accuracy varies accordingly.

IndoTrack, by Li et al. (2017), is the foundational device-free CSI tracking paper commonly cited in this space. More recent work narrows toward this project's use case: a fingerprinting approach by Jiang and Yuan (2025) combines a Gaussian Mixture Model with weighted k-NN and Kalman filter tracking for device-free positioning. Shen et al.'s time-selective RNN approach (2023) directly addresses the busy-room, multi-person setting, and Chung et al.'s WiRiS (2023) uses a transformer architecture over RIS-assisted CSI to jointly solve people counting and localisation, which is the closest single-paper match to "how many people, where" as a joint problem.

Latency here depends heavily on the specific access point, not just the sensing algorithm: Shen et al. (2023) report a CSI sampling rate of 10 Hz, explicitly capped by their TP-Link WDR4300 802.11n router rather than by their model, whereas IndoTrack is reported in secondary sources to have used a considerably faster 100 Hz commodity CSI stream, a figure this chapter could not independently confirm against the primary paper. Chung et al. (2023) do not report any update rate or latency figure at all, evaluating WiRiS purely on counting/localisation accuracy. This order-of-magnitude spread (10 Hz to 100 Hz) across otherwise comparable commodity Wi-Fi setups is itself a latency risk to flag for this criterion, separate from the modality's already-noted lack of height output.

Implication for RQ1: none of the gathered Wi-Fi CSI sources report height or pose-level output, and the modality's environment-sensitive spatial resolution limits its usefulness for producing a height estimate on its own. It remains worth carrying forward as a low-cost option for presence/room-occupancy sensing.

Pros:

- Effectively free: reuses existing Wi-Fi infrastructure rather than needing dedicated hardware.
- Multi-person and multi-room literature exists (Time-Selective RNN, WiRiS).
- Device-free output with no recognisable imagery.

Cons:

- No published height-estimation precedent at all in the literature gathered for this chapter, the single biggest gap against RQ1.
- More sensitive to environmental changes (furniture, multipath) than radar or thermal.
- No demonstrated real-time embedded deployment in the sources reviewed.

Ultrasonic Ranging

Active ultrasonic ranging measures distance via time-difference-of-arrival (TDOA) or time-of-flight on emitted acoustic pulses, rather than passively listening for ambient sound. Because the system controls its own signal, this sidesteps the "room full of talking people" interference problem that a passive-listening approach would face.

Caicedo and Pandharipande (2012) use a beamforming ultrasonic array with moving-target-indicator processing for indoor presence detection, and report it outperforming PIR sensors for occupancy detection. Ranging resolution can be tight: Segers et al. (2015) use frequency-shift-keyed ultrasound to track up to four simultaneous transmitters at 1-2cm accuracy on an FPGA, and Holm and Nilsen (2010) demonstrate TDOA-based positioning from a transmitter array, a useful reference for multi-receiver node geometry if this technology is pursued further. Wang et al. (2017) fuse an ultrasonic array with vision for 3D human tracking, which is relevant mainly as a sensor-fusion architecture reference given this project's non-camera constraint.

Neither Segers et al. (2015) nor Caicedo and Pandharipande (2012) report an update rate,

refresh rate, or end-to-end latency figure for their systems, only accuracy and, for Segers et al., the underlying ADC sampling rate used internally for correlation. Latency for this modality is therefore an open question rather than a documented figure, an additional gap to close in any dedicated bench-testing pass.

Implication for RQ1: ultrasonic ranging gives decent range accuracy via TDOA/ToF, but azimuth resolution is limited, and full 2D/3D coverage needs multiple synchronised nodes, introducing multi-sensor complexity that using active signals was otherwise meant to avoid. No gathered source addresses height or elevation for ultrasonic ranging. A vertically-oriented array would be needed to get a z-axis reading at all, which is not demonstrated in the current evidence base.

Pros:

- Active-signal approach sidesteps the "room full of talking people" interference problem passive listening would face.
- Demonstrated real-time operation on embedded hardware (an FPGA, per Segers et al.).
- Reasonable literature base for ranging and positioning.

Cons:

- No gathered source addresses height or elevation for ultrasonic ranging.
- Coarse azimuth resolution compared to radar.
- Full coverage needs multiple synchronised nodes, reintroducing multi-sensor complexity.
- One of the pricier custom-built options (€200-400).

Infrared / Thermal

Low-resolution thermal/thermopile arrays measure a grid of per-pixel temperature readings. A person would appear as a warm blob against the usually cooler background. Resolution and range are both limited, which constrains this technology to smaller rooms, but the low pixel count is itself a privacy advantage. The output is far below the resolution needed to identify a person or resolve recognisable imagery, which is a strong fit for the sub-question on minimising personally identifiable data from child visitors.

Most directly relevant to RQ1, Naser et al. (2021) explicitly extend thermal-array distance estimation to height measurement using a neural network. This directly targets height estimation in the same way as this project's research question, and should be read first. Feasibility on embedded hardware is demonstrated by Shubha and Shastrimath (2022), who report 99% occupancy accuracy using a Grid-EYE/AMG8833 thermopile array on a Raspberry Pi, addressing RQ2's embedded-platform requirement directly. Foundational Grid-EYE-based detection and tracking work by Shetty et al. (2017) is extended by Klir et al. (2025), who apply YOLOv5/DETR-style detection to 32x32 thermal images for multi-person localisation with a generative data model, which is relevant to the multi-person extensibility sub-question (RQ4). Detection robustness under changing conditions is addressed by Trofimova et al. (2017), who use adaptive background estimation for thermal-array-based human detection.

On latency, Naser et al. (2021) report the thermal sensor used in their height-estimation work supports a configurable sampling rate between 0.5 Hz and 64 Hz. The specific Grid-EYE/AMG8833 breakout identified below for procurement is a simpler, coarser part by comparison: its datasheet exposes only a 10 Hz or 1 Hz frame-rate register, not the wider range Naser et al.'s sensor supports. Shubha and Shastrimath (2022) are reported in secondary sources to have run their AMG8833/Raspberry Pi occupancy pipeline at 10 Hz, consistent with the datasheet's faster setting, though this could not be independently confirmed against the primary paper.

Implication for RQ1: thermal array has a source explicitly performing height estimation from sensor data, and its low resolution doubles as a privacy strength rather than only a limitation. The main tradeoff to carry forward is its effective range, which needs to be checked against the actual dimensions of the interactive space.

Sensor hardware cost is substantially lower than mmWave, in the order of €40-55 excluding the embedded compute platform, based on the [Adafruit AMG8833 breakout](#) list price, and the sensor interface (I2C) is simpler to bring up than a radar UART/configuration pipeline.

Pros:

- The only technology in this review with a source performing height estimation directly from sensor data (Naser et al.).
- Cheapest viable sensor-only option found (€40-55).
- Demonstrated real-time embedded operation on a Raspberry Pi (Shubha and Shastrimath).
- Low pixel count doubles as a privacy strength: the output cannot resolve recognisable imagery.
- Simpler sensor interface (I2C) than a radar UART/configuration pipeline.

Cons:

- Shorter effective range than mmWave, constraining it to smaller rooms.
- Lower spatial resolution overall, which needs checking against the Holobox's actual dimensions.

Capacitive & Floor-based Sensing

Capacitive / Electric-Field Sensing

Two device-free variants exist: proximity-plate sensing, which measures distance to a body via a change in capacitance, and floor-embedded capacitive electrodes, which require a person to be standing on the sensed area.

Ramezani Akhmareh et al. (2016) offer a good starting point for this sensing family, with a clear related-work review of alternative capacitive designs. Faulkner et al.'s CapLoc (2020) is a floor-embedded capacitive sensing mat, building on the earlier CapFloor system, evaluated in the EvAAL indoor localisation competition for device-free localisation and fall detection. The

broadest source is Bian et al.'s survey (2024), covering proximity capacitive, floor capacitive, and passive electric-field sensing side by side. A passive variant, Fu et al.'s Linoc system (2017), is reported as less noise-prone than active capacitive sensing, worth considering if EMI is a concern in the interaction space.

Range is the main practical constraint: plate sensors typically top out around one to two electrode diameters (roughly 0.3-1m), so whole-room coverage needs either a floor grid or many wall/ceiling nodes, good for near-field precision, not for single-sensor room-scale coverage. The signal is also inherently a presence/proximity measure rather than a spatial one, so it is not naturally suited to a height reading at all.

Implication for RQ1: capacitive sensing has no reported height output, and its short range makes whole-room coverage costly without a dense sensor grid.

Floor-based (Pressure) Sensing

Floor-based sensing (pressure, piezoresistive, or capacitive-floor mats) is inherently a 2D floor-plane measurement, but because each footstep is a spatially discrete event, it naturally disambiguates multiple people.

Gatto et al. (2015) use a 64x64 pressure-switch array per floor tile to track both humans and robots. Al-Naimi et al. (2014) fuse pressure-floor tracking with pyroelectric infrared sensors, and report the combination keeps working when individual people become hard to distinguish by pressure alone. Feng et al. (2015) use a fiber-optic floor grid explicitly built for multi-target localisation using group-testing theory, and Ibara et al. (2013) show gait-based individual identification from stride length and direction, noting extensibility to multi-person tracking.

Implication for RQ1: floor sensing's real strength is that it trivially solves "how many people, where" in a crowded room, since footsteps are naturally separated events. Its weaknesses are installation cost (it has to go under the actual floor) and a complete absence of height information. Carrying this forward would mean pairing it with a separate height-sensing modality purely for the z-axis.

Pros:

- Floor-based variants trivially solve "how many people, where," since footsteps are naturally separated events.
- Solid literature base for both the capacitive and floor-pressure sub-variants.
- Inherently non-identifying (a presence/proximity or pressure signal, not imagery).

Cons:

- No height signal at all from either sub-variant.
- Capacitive sensing has short range, needing a dense grid for whole-room coverage.
- Floor-based sensing requires physical installation under the existing floor.
- Cost is highly variable and, for floor installations, can run into the thousands of euros.

Ultra-Wideband (UWB)

Device-free UWB sensing (IR-UWB) uses very short radio pulses across a wide bandwidth to estimate range and, via signal-processing or ML-based analysis of the reflected signal, detect and count people without requiring them to carry a tag. Evidence gathered so far is limited to a single source: Njanda et al. (2024) apply machine learning to IR-UWB radar for people counting.

The representative module named for this approach, the Novelda Xethru X4, does not have a single fixed frame rate in its own datasheet: frame rate directly trades off against SNR/processing gain (up to a maximum pulse repetition frequency of 40.5 MHz), so any specific frames-per-second figure is an application-specific configuration choice, not a hardware spec. Njanda et al.'s paper itself could not be confirmed to report a frame rate or the specific radar hardware used.

Implication for RQ1: the evidence base for UWB is currently too thin to draw a conclusion for this project as no gathered source addresses height or pose-level output for UWB. Given that UWB ranging chips are commonly cheap and embedded-friendly, this is flagged for a further literature pass before being ruled in or out, rather than being ruled out on the current single source.

Pros:

- People-counting literature exists (Njanda et al., reporting 99% accuracy with an ANN classifier).
- Radar-based output is inherently non-identifying.

Cons:

- No height or pose-level literature at all; flagged in this review as unproven.
- Only a single source exists, described in this chapter as too thin an evidence base to draw a conclusion from.
- The representative module identified (Novelda Xethru X4) is no longer available through an authorised distributor, making cost and sourcing unverifiable.

2.4.2. Multi-Person: Identifying the Active Speaker

Everything above addresses single-person position and height. A separate problem, relevant to RQ4's extensibility to multi-person scenarios, is that once several people are present in the interactive space, the system also needs to determine which one is currently speaking to the Virtual Human, without a camera, and without relying on people carrying a microphone or tag.

mmWave Radar: Resolution Limits for Counting, Height, and Speaker Separation

Whether the IWR6843 reports two people as distinct targets, rather than merging them into one detection, depends on the radar's angular and range resolution, not on a single blanket "multi-person" capability. Two people need roughly 1.7-2.9m of horizontal (left-right) separation, or a height difference on the order of 1.7m (e.g. one seated, one standing), before the radar's angular

resolution reliably reports them as two separate people instead of one merged detection. Range separation (front-back, i.e. distance from the sensor) is far finer, on the order of 4-5cm, and follows automatically from the radar's time-of-flight measurement rather than needing angular discrimination, so it is not a practical constraint in the way horizontal or height separation is.

Speaker identification does not share this constraint. It relies on Doppler micro-motion detection, the small, characteristic mouth/chest movements produced during speech (the same underlying mechanism as the mmWave vibrometry work cited below), rather than on spatial separation between people. This means the radar can identify who is speaking regardless of how close together multiple people are standing, even in cases where their positions are too close to separate cleanly by angle.

For a room roughly 5m wide with 2-3 people scattered across it, ordinary spacing is sufficient to satisfy all three requirements together: counting people, estimating each person's height, and identifying the active speaker. The failure case to plan around is 4 or more people clustered tightly together, where horizontal separation drops below the 1.7-2.9m angular-resolution threshold and individual tracks become ambiguous, the same underlying failure mode as the "crossover ambiguity" already noted for 2D LiDAR above, rather than a new risk specific to mmWave.

mmWave Vibrometry

mmWave radar's vibrometry capability, established above for position and pose sensing, extends to sound itself: Ozturk et al. (2021) show that mmWave radar can reconstruct audio from the surface vibrations of a sound source, including a human throat, working even through soundproofing. More directly on point, Xu et al. (2019) build WaveEar, a system that uses a low-cost mmWave probe to first localise the position of the speaker among multiple people, then steer the radar toward that person's throat to sense their vocal vibrations. This effectively solves "who is speaking" and "what are they saying" from the same non-camera sensor.

Implication for RQ4: mmWave vibrometry can localise a speaker among multiple people and recover their voice from the same radar signal. Neither source has been tested in this project's specific context (children's voices, a room-scale interactive installation), so this is flagged for a dedicated literature and bench-testing pass.

Wi-Fi CSI Sensing

Wi-Fi CSI reflections from mouth and lip movement during speech are detectable without any additional hardware beyond commodity Wi-Fi. Wang et al.'s WiHear (2014) uses directional antennas aimed at a target's face to detect fine-grained radio reflections from mouth movement, and reports being able to sense multiple people's talk simultaneously using MIMO. Du et al.'s WiTalk (2018) reports over 92% recognition accuracy discerning a set of syllables from WiFi CSI dynamics alone.

Implication for RQ4: Wi-Fi CSI can detect and analyse mouth-movement reflections associated with speech using only commodity Wi-Fi hardware. Neither source has been tested in this project's specific context (children's voices, a room-scale interactive installation), so this is

flagged for a dedicated literature and bench-testing pass.

UWB Radar

IR-UWB radar can remotely detect motion of the lips and jaw associated with speech. Shin and Seo (2016) extract lip and jaw motion features from IR-UWB radar signals for contactless silent speech recognition, improving recognition performance over using raw radar signal features directly.

Implication for RQ4: IR-UWB radar can detect articulator motion associated with speech from radar reflections alone. This source has not been tested in this project's specific context (children's voices, a room-scale interactive installation), so this is flagged for a dedicated literature and bench-testing pass.

Laser Doppler Vibrometry

A laser Doppler vibrometer (LDV) measures skin-surface vibration at a distance via the Doppler shift of a reflected laser beam, and can pick up throat vibration during speech even in acoustically noisy conditions. Avargel and Cohen (2011) combine an LDV sensor aimed at the larynx with a standard microphone, using a pre-filtering and voice-activity-detection algorithm to enhance speech quality under transient noise.

Implication for RQ4: LDV can pick up a specific person's throat vibration with high immunity to ambient acoustic noise, but it requires the laser to be aimed precisely at that person's throat, and several published LDV setups rely on a camera to aim and track the target, which would need a non-camera aiming mechanism to fit this project's constraints. This source has not been tested in this project's specific context (children's voices, a room-scale interactive installation), so this is flagged for a dedicated literature and bench-testing pass.

Microphone Array (Passive Acoustic)

Berghi and Jackson (2023) use a microphone array with a convolutional recurrent neural network to detect and localise an active talker's horizontal position from multichannel audio alone, comparing GCC-PHAT, SALSA features, and beamforming as input representations. This requires no camera or radar hardware at all, only an array of microphones placed in the interactive space.

Implication for RQ4: a microphone array can localise an active talker's position from audio alone, with no line-of-sight or radar hardware requirement. It has not been tested in this project's specific context (children's voices, a room-scale interactive installation), so this is flagged for a dedicated literature and bench-testing pass.

2.4.3. Combined Sensor Approaches

The individual technologies above are also weighed as sensor combinations, where pairing two of them addresses a gap neither has alone.

mmWave + Thermal Fusion for Height

The project's own risk analysis anticipates that a single sensor may not reliably discriminate height across the full range, and names combining "two or more complementary sensors (fusion)" as the planned fallback. The concrete instantiation of that plan is to add the infrared/thermal array as a second, height-specific sensor alongside mmWave, rather than replacing mmWave outright: mmWave continues to provide position and person tracking, while the thermal array's independent height reading, backed by Naser et al. (2021), is used to confirm, or substitute for, mmWave's own height estimate.

This is a narrower, lower-risk form of fusion than **2D LiDAR + Thermal Fusion** below: that combination requires fusing two independent position-sensing streams from scratch, whereas this option keeps mmWave as the sole position sensor and only adds a second signal for the specific sub-problem, height. The main added implementation work is registering the thermal array's field of view against the mmWave-detected person's position, rather than fusing two full tracking pipelines. Combined sensor-only hardware cost is the sum of the mmWave and thermal entries in **Section 2.4.4**, roughly €200-240, plus a single shared compute platform.

Pros:

- Combines mmWave's strong position/multi-person tracking with thermal's direct height-estimation precedent.
- Gives a concrete fallback if mmWave's own height estimate underperforms, without discarding the mmWave investment.
- Reuses the same thermal hardware already procured in the prototyping plan's first phase, rather than adding a third sensor.

Cons:

- Requires registering the thermal array's field of view against the mmWave-detected person's position: real added implementation work.
- Combined cost still includes the pricier mmWave component (€200-240 total).
- No single published source tests this exact combination; each half is validated separately, not together.

Budget Tier: HLK-LD2450 + Thermal Fusion

A further budget tier exists by swapping the position sensor itself. The **HLK-LD2450** is a 24 GHz radar module, around €8-12, that tracks X/Y position and speed for up to three people simultaneously over UART, with a 6 m range and a 10 Hz update rate. Paired with the same infrared/thermal array for height, following the same fusion pattern as above, combined sensor-only hardware cost drops to roughly €50-70, a fraction of either mmWave-based option.

This option carries a materially different kind of risk than the rest of this chapter. Every other candidate technology evaluated here is backed by at least one peer-reviewed source; no such source was found for the HLK-LD2450. It is documented only through the manufacturer's own

manual and hobbyist/smart-home community projects (e.g. Home Assistant and ESPHome integrations), not academic literature. Choosing it would mean the position-tracking half of the system rests entirely on this project's own bench validation rather than on prior published results. That is a different kind of unknown than "will the accuracy be good enough": it is "has anyone shown this works for this purpose at all." Like 2D LiDAR, it also only outputs a single scan-plane (x, y) position, so it inherits the same reliance on a second sensor for height.

Pros:

- Dramatically cheaper than either mmWave-based option (€50-70 total, sensor-only).
- Still gets a height reading via the same literature-backed thermal array.
- The HLK-LD2450 already tracks up to three people's X/Y position and speed out of the box.

Cons:

- No peer-reviewed source exists for the HLK-LD2450; it is documented only through the manufacturer's manual and hobbyist community projects.
- The position-tracking half of the system would rest entirely on this project's own bench validation, not prior published results.
- Only proposed as a fallback of last resort, gated on a standalone feasibility test before being trusted.

2D LiDAR + Thermal Fusion

Pros:

- 2D LiDAR contributes strong in-plane position accuracy and occlusion tolerance (Bouazizi et al.).
- Thermal contributes the same direct height-estimation precedent used elsewhere in this chapter (Naser et al.).

Cons:

- The most expensive combination reviewed in this chapter (€290-345, sensor-only).
- Combining two independent position-sensing streams from scratch introduces synchronisation and fusion complexity flagged as a disproportionate implementation risk for this project's timeline, per [Timeline](#).
- Unlike the mmWave + Thermal option, this does not reuse an already-working single-sensor pipeline.

2.4.4. Comparison Summary

Table 3. Candidate Sensor Comparison Summary

Technology	Representative Module	Approx. Hardware Cost (sensor only)	Height Data in Literature	Documented Latency / Update Rate
mmWave Radar	TI IWR6843AOPEVM (see mmWave Module Selection for the variant comparison)	€160-185	Yes, keypoint-based	IWR6843 datasheet: up to ~76 Hz, typically 10-20 Hz in practice. Cited tracking papers report 15-20 fps on different radar chips.
Infrared/Thermal Array	Adafruit AMG8833 breakout	€40-55	Yes, direct	AMG8833 datasheet: 10 Hz or 1 Hz selectable.
2D LiDAR + Thermal (sensor fusion)	RPLIDAR A2M8 + AMG8833	€290-345	Partial, via fusion	RPLIDAR A2M8 datasheet: 5-15 Hz scan rate, typically 10 Hz.
Wi-Fi CSI Sensing	None. It reuses existing commodity Wi-Fi hardware	Approx. €0 (existing infrastructure)	None found	Hardware-dependent: 10 Hz (Shen et al., 2023) up to a reported 100 Hz (IndoTrack, unverified against primary source).
Ultrasonic Ranging	Custom transducer array (e.g. SPM0404UD5 MEMS elements, per Segers et al. (2015))	€200-400 (rough estimate for custom-built hardware, not a single purchasable SKU)	None found	Not reported in gathered sources.
Capacitive & Floor-based Sensing	Custom-built capacitive/pressure floor mat. There is no off-the-shelf module	€80-2000+, depending on variant (rough estimate for custom-built hardware)	None found	Not reported in gathered sources.

Technology	Representative Module	Approx. Hardware Cost (sensor only)	Height Data in Literature	Documented Latency / Update Rate
Ultra-Wideband (UWB)	Novelda Xethru X4 (per Njanda et al. (2024))	Unverifiable: no longer available through an authorised distributor; only found via secondary-market listings, so not treated as a sourceable cost	Unproven	Not a fixed spec: trades off against SNR, up to a 40.5 MHz pulse repetition frequency ceiling per the Xethru X4 datasheet.

Figures above are sensor/module hardware only. All candidates other than Wi-Fi CSI also need an embedded compute platform to run the sensing pipeline; that cost is excluded here since it is a shared requirement across candidates rather than something that distinguishes one sensor choice from another. Latency figures are the sensor’s own documented output rate, not a measured end-to-end pipeline latency; see the caveat already noted in Evaluation Criteria.

NOTE

No project budget has yet been agreed with MindLabs, so the cost figures above are indicative market prices, not an approved spend. mmWave is, by a wide margin, the most expensive of the two technically strongest candidates (roughly three to four times the thermal array’s sensor cost), which is why cost, not just literature strength, has to be weighed explicitly in the Recommended Approach below.

Table 4. Criteria Comparison Matrix

Technology	Height Est.	Embedded Real-Time	Multi-Person	Fusion-Ready	Privacy	Literature	Affordable
mmWave Radar	✓	✓	✓	✓	✓	✓	☐
Infrared/Thermal Array	✓	✓	✓	✓	✓	✓	✓
mmWave + Thermal Fusion	✓	✓	✓	✓	✓	☐	☐
Budget Tier (HLK-LD2450 + Thermal)	✓	✓	✓	✓	✓	☐	✓
2D LiDAR + Thermal Fusion	✓	☐	✓	✓	✓	✓	☐
Wi-Fi CSI Sensing	☐	☐	✓	☐	✓	✓	✓
Ultrasonic Ranging	☐	✓	☐	☐	✓	✓	☐

Technology	Height Est.	Embedded Real-Time	Multi-Person	Fusion-Ready	Privacy	Literature	Affordable
Capacitive & Floor-based Sensing	☐	☐	✓	☐	✓	✓	☐
Ultra-Wideband (UWB)	☐	☐	✓	☐	✓	☐	☐

NOTE

Marks reflect what the literature and product analysis gathered in this chapter actually establish, not an abstract technical ceiling. A cross means the gathered sources do not demonstrate that capability for this specific approach, not that it is proven impossible. "Literature" is marked against each approach as a whole: the two fusion options each combine two individually well-documented sensors, but no published source tests either combination together, which is why they cross where their individual components would not.

2.4.5. Recommended Approach: mmWave, Phased by Capability

On the criteria set out earlier, mmWave radar comes out ahead: it is the only approach with keypoint-level pose literature, it extends to the RQ4 speaker-identification problem via the same sensor, its output is inherently non-identifying, and it has the strongest literature base of any candidate reviewed. The infrared/thermal array remains the only approach with a direct height-estimation precedent and by far the cheapest to acquire, but on technical merit mmWave is the stronger choice, and it is recommended directly rather than deferred behind a cheaper sensor. This is a recommendation of the mmWave modality; which specific IWR6843 evaluation-board variant to build with is a separate, narrower decision, addressed next in [mmWave Module Selection](#).

mmWave is, by a wide margin, still the most expensive candidate reviewed (roughly three to four times the thermal array's sensor cost), and no project budget has yet been agreed with MindLabs. That cost is treated as a precondition to clear with the company mentor before hardware is ordered, not as a reason to build something else first. What is staged instead is the implementation itself: rather than assuming position tracking, height estimation, multi-person clustering, and speaker detection all work together on the first attempt, the mmWave pipeline is built up one capability at a time, each on top of an already-working previous phase. This still isolates risk (a problem in one capability does not require redoing the whole pipeline) without introducing a second sensor and a mid-project hardware swap. If mmWave's own height estimate underperforms once measured, [mmWave + Thermal Fusion for Height](#) remains the documented fallback; if the mmWave budget cannot be approved at all, [the HLK-LD2450 budget tier](#) is the fallback of last resort.

The concrete phasing of this capability build-up is carried into the Conclusion and the Prototyping Plan chapter.

2.5. mmWave Module Selection

Candidate Sensor Technologies recommends mmWave radar as the sensing modality, built around the TI IWR6843 chip. That chip ships on several evaluation-board variants at different price points and field-of-view characteristics, given below as azimuth (angular coverage side-to-side) / elevation (angular coverage top-to-bottom, i.e. floor-to-head reach), all sharing the same TI mmWave SDK/point-cloud software stack and all operable standalone over USB (none strictly require the separate MMWAVEICBOOST carrier card for point-cloud demos). This chapter compares those variants directly and picks one to carry into the Prototyping Plan.

- **IWR6843LEVM**: TI's cost-optimised FR4-antenna variant, €140-160 at current [Digi-Key](#) listings, with a 120° azimuth / 80° elevation field of view. The cheapest usable (i.e. not bare-chip) option found.
- **IWR6843ISK**: the standard long-range antenna plug-in module, €175-265 at current [Mouser](#) listings, with a narrower 108° azimuth / 44° elevation field of view tuned for range over coverage. Better documented in some community projects, but its narrow elevation beam works against a room-scale height-estimation use case rather than for it.
- **IWR6843ISK-ODS**: a wide-field-of-view (120° azimuth / 120° elevation) antenna variant, etched on a Rogers substrate, that TI markets specifically for overhead/ceiling-mounted people counting and tracking, €195-215, ~12 m range. Better suited than the plain ISK if the Holobox sensor ends up ceiling-mounted, since that use case is exactly what this variant's antenna is designed for.
- **IWR6843AOPEVM**: an antenna-on-package (AoP) variant, with the antenna integrated directly into the chip package rather than etched onto a separate PCB, €160-185 at current [Mouser](#) / [Digi-Key](#) listings. Matches the ISK-ODS's wide 120° azimuth / 120° elevation field of view in a smaller (~65×60mm), standard (non-overhead-specific) form factor, at roughly the same price as the plain LEVM.
- **Infineon BGT60TR13C** eval board: a comparable, EU-sourced 60 GHz alternative outside the TI ecosystem, €230.

Table 5. mmWave EVM Variant Comparison

Module	Antenna	Field of View (Az/EI)	Approx. Price	Best fit
IWR6843LEVM	Cost-optimised FR4 PCB antenna	120°/80°	€140-160	Cheapest usable option; adequate but not class-leading elevation coverage.
IWR6843ISK	Long-range PCB antenna	108°/44°	€175-265	Narrow elevation FoV suits a defined industrial zone, not room-scale height coverage.

Module	Antenna	Field of View (Az/El)	Approx. Price	Best fit
IWR6843ISK-ODS	Wide-FoV PCB antenna (Rogers substrate)	120°/120°	€195-215	Purpose-built for ceiling/overhead mounting specifically.
IWR6843AOPEVM	Antenna-on-package (integrated into chip)	120°/120°	€160-185	Same wide coverage as ISK-ODS, more compact, mount-agnostic, and no external antenna to design around.
Infineon BGT60TR13C	PCB antenna (EU-sourced)	Not independently verified	€230	Non-TI fallback if TI sourcing becomes a problem.

On field of view, the ISK's narrow 44° elevation beam is a poor match for this project: it is tuned for a defined industrial zone at range, not for seeing a person from roughly floor to head height across a room, which is what the height-estimation use case actually needs. The ISK-ODS and AOPEVM both offer the wide 120° elevation FoV that fits room-scale height coverage, at similar prices (€160-185 to €195-215) and near the LEVM's own price point.

2.5.1. Recommendation: IWR6843AOPEVM

Between the ISK-ODS and the AOPEVM, this chapter recommends the **IWR6843AOPEVM**: it matches the ISK-ODS's field of view without being purpose-tuned to overhead-only mounting (leaving mounting position flexible while the Holobox's sensor placement is still being finalised), its antenna-on-package design removes the external antenna as a separate design/tuning problem entirely (a real simplification for a small embedded team, and a lower-risk path if the module is ever redesigned onto a custom production board), and its more compact ~65×60mm footprint eases fitting inside the Holobox's existing enclosure, per the power-consumption-and-footprint criterion in Evaluation Criteria. The trade-off is that a 120° elevation beam picks up more off-axis/peripheral reflections than the ISK's narrower beam, which the clutter-removal and clustering step (following Huang et al.) already in the pipeline is expected to absorb, so it is not treated as disqualifying.

This module choice does not change the mmWave modality recommendation or its cost bracket (€160-185 sensor-only, in the same range as the plain LEVM), so the phased-build plan and fallback options set out in Candidate Sensor Technologies' Recommended Approach carry forward unchanged.

2.6. Conclusion

Candidate Sensor Technologies weighed the technically viable sensors and sensor combinations against the criteria in Evaluation Criteria. On that basis, mmWave radar is recommended directly: it is the only approach with keypoint-level pose literature for height estimation, it extends to the RQ4 active-speaker problem via the same sensor, its output is inherently non-identifying, and it has the strongest literature base of any candidate reviewed. Within the TI IWR6843 family, mmWave Module Selection picks the **IWR6843AOPEVM**: its antenna-on-package design gives the wide 120° elevation field of view needed for room-scale height coverage without a separate antenna to design around, in a compact, mount-agnostic form factor at roughly the same price as the cheaper plain LEVM variant.

Rather than deferring mmWave behind a cheaper sensor, implementation risk is managed by splitting the mmWave build itself into capability-based phases: position tracking first, then height estimation, then multi-person clustering, then speaker detection, each building on an already-working previous phase. This still isolates risk (a problem in one capability does not require redoing the whole pipeline) without introducing a second sensor and a mid-project hardware swap. mmWave remains, by a wide margin, the most expensive candidate reviewed, and no project budget has yet been agreed with MindLabs, so mentor approval of the full mmWave hardware cost is a precondition for this plan, not a detail to resolve later.

The concrete phasing of this recommendation is set out in the Prototyping Plan chapter.

2.7. Prototyping Plan

The Conclusion recommends mmWave radar directly, built up in capability-based phases rather than deferred behind a cheaper sensor. This chapter carries that recommendation into a concrete plan for the bench-experimentation stage described in Research Approach.

2.7.1. Approach

Build the mmWave pipeline once, adding one capability at a time on top of an already-working previous phase, rather than assuming position tracking, height estimation, multi-person clustering, and speaker detection all work together on the first attempt.

Phase 1: Position Tracking

Bring up the core sensor-to-output pipeline before any height, multi-person, or speaker-detection logic is added:

- mmWave sensor driver: UART point-cloud parser for the IWR6843AOPEVM.
- Clutter removal, clustering, and Kalman filtering for position tracking, following Huang et al. (2021).
- An Unreal Engine socket bridge (C++ TCP output).

This phase validates the parts of the system every later phase depends on: sensor bring-up, point-cloud parsing, and the Unreal integration. A failure here is a failure in the foundation, so it is deliberately tackled before any of the more research-heavy capabilities below.

Phase 2: Height Estimation

Extend the working (x, y) position tuple from Phase 1 to (x, y, z):

- Choose between a lightweight blob-size heuristic and a keypoint-extraction model following ProbRadarM3F-style approaches (2024) and Zheng et al. (2025), trading implementation effort against accuracy.
- Feed the resulting height value into the same output tuple and Unreal bridge already validated in Phase 1.

Fallback: if mmWave's own height estimate underperforms once measured, mmWave + Thermal Fusion for Height is the documented next step, adding a thermal array as a second, height-specific sensor rather than replacing mmWave.

Phase 3: Multi-Person Clustering

Extend the Phase 1 clustering step to reliably separate multiple simultaneous people in the point cloud, rather than assuming a single occupant, following Huang et al.'s (2021) multi-person tracking demonstration. This validates RQ4's extensibility requirement at the architecture level before speaker detection is added on top of it.

Phase 4: Speaker Detection

Add mmWave vibrometry to localise and steer toward an active speaker's throat, reusing the same radar hardware and multi-person clustering from Phase 3, following Xu et al.'s WaveEar (2019). This completes RQ4's active-speaker identification requirement using the same sensor family throughout, rather than a second, dedicated device.

Risk mitigation: because each phase adds a capability on top of an already-working previous phase rather than swapping hardware, a problem in one phase is isolated to that phase; if Phase 4's speaker detection proves difficult, Phases 1-3 remain a working position/height/multi-person system rather than a dead end. If the mmWave budget cannot be approved at all, the HLK-LD2450 + thermal budget tier is the fallback of last resort, gated on its own short bench test since no published source can de-risk it in advance.

NOTE

Phase durations are not fixed here; they should be mapped onto the sprint structure in [Timeline](#), since that document is the authoritative one for actual dates.

2.7.2. Next Steps

- Confirm the physical dimensions of the Holobox interaction space, since this determines the effective range required from the mmWave sensor.
- Present this plan and the mmWave hardware cost estimate to the company mentor for budget approval, per the process in [Finance and Risks](#); no hardware is ordered before that approval.
- Once approved, procure the mmWave module and begin bench experimentation with Phase 1, as described earlier.
- Treat Phase 4's speaker detection as a follow-on capability to validate once Phases 1-3 are working, rather than a day-one requirement.

3. Bibliography

- Alam, M. A. U., Rahman, M. M., & Widberg, J. Q. (2021). *PALMAR: Towards Adaptive Multi-inhabitant Activity Recognition in Point-Cloud Technology*. <https://arxiv.org/pdf/2106.11902>
- Almalioglu, Y., Turan, M., Lu, C. X., Trigoni, N., & Markham, A. (2019). *Milli-RIO: Ego-Motion Estimation with Low-Cost Millimetre-Wave Radar*. <https://arxiv.org/pdf/1909.05774>
- Al-Naimi, I., Wong, C. B., Moore, P., & Chen, X. (2014). *Advanced approach for indoor identification and tracking using smart floor and pyroelectric infrared sensors*. 5th International Conference on Information and Communication Systems (ICICS), Irbid, Jordan. <https://www.researchgate.net/publication/271463149>
- Avargel, Y., & Cohen, I. (2011). *Speech measurements using a laser Doppler vibrometer sensor: Application to speech enhancement*. Joint Workshop on Hands-free Speech Communication and Microphone Arrays (HSCMA), Edinburgh, Scotland. <https://israelcohen.com/wp-content/uploads/2018/05/HSCMA2011.pdf>
- Berghi, D., & Jackson, P. J. B. (2023). *Audio Inputs for Active Speaker Detection and Localization via Microphone Array*. <https://arxiv.org/pdf/2307.14739>
- Bian, S., Liu, M., Zhou, B., Lukowicz, P., & Magno, M. (2024). *Body-Area Capacitive or Electric Field Sensing for Human Activity Recognition and Human-Computer Interaction: A Comprehensive Survey*. arXiv. <https://arxiv.org/pdf/2401.06000>
- Bouazizi, M., Mora, A. L., Feghoul, K., & Ohtsuki, T. (2024). *Activity Detection in Indoor Environments Using Multiple 2D Lidars*. Sensors. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11154263/>
- Caicedo, D., & Pandharipande, A. (2012). *Ultrasonic array sensor for indoor presence detection*. 20th European Signal Processing Conference (EUSIPCO), Bucharest, Romania. <https://www.semanticscholar.org/paper/3bd6ffc871056616ba364d8f10f6d3904df3a487>
- Chung, W.-Y., Shen, L.-H., Feng, K.-T., Lin, Y.-C., Lin, S.-C., & Chang, S.-F. (2023). *WiRiS: Transformer for RIS-Assisted Device-Free Sensing for Joint People Counting and Localization using Wi-Fi CSI*. <https://arxiv.org/pdf/2304.06475>
- Du, C., Yuan, X., Lou, W., & Hou, Y. T. (2018). *Context-Free Fine-Grained Motion Sensing Using WiFi (WiTalk)*. IEEE International Conference on Sensing, Communication, and Networking (SECON). https://www.cnsr.ictas.vt.edu/publication/secon18_final.pdf
- Enoki, M., Watanabe, K., & Noguchi, H. (2024). *Single Person Identification and Activity Estimation in a Room from Waist-Level Contours Captured by 2D Light Detection and Ranging*. Sensors, 24(4), 1272. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10892201/>
- Faulkner, N., Parr, B., Alam, F., Legg, M., & Demidenko, S. (2020). *Device Free Localization with Capacitive Sensing Floor (CapLoc)*. IEEE Sensors Applications Symposium (SAS). <https://www.researchgate.net/publication/344620665>

Builds on the earlier CapFloor system by Braun, A., Heggen, H., & Wichert, R. (2011), presented at the EvAAL indoor localisation competition.

Feng, G., Yang, Y., Guo, X., & Wang, G. (2015). *A Smart Fiber Floor for Indoor Target Localization*. IEEE Pervasive Computing, 14(2), 52-59. <https://www.researchgate.net/publication/273394655>

Fu, B., Kirchbuchner, F., von Wilmsdorff, J., Grosse-Puppenthal, T., Braun, A., & Kuijper, A. (2017). *Indoor Localization Based on Passive Electric Field Sensing* (the Linoc system). Ambient Intelligence (AmI 2017), Lecture Notes in Computer Science vol. 10217, 64-79. <https://www.researchgate.net/publication/315799691>

Gatto, A., Bassoli, E., Contigiani, M., Frontoni, E., Mancini, A., & Zingaretti, P. (2015). *Energy Harvesting Smart Floor for Indoor People Localization and Tracking*. ASME International Design Engineering Technical Conferences (IDETC-CIE). <https://www.researchgate.net/publication/314616906>

Holm, S., & Nilsen, C.-I. C. (2010). *Robust ultrasonic indoor positioning using transmitter arrays*. Indoor Positioning and Indoor Navigation (IPIN) 2010. <https://www.researchgate.net/publication/224198462>

Huang, X., Cheena, H., Thomas, A., & Tsoi, J. K. P. (2021). *Indoor Detection and Tracking of People Using mmWave Radar*. Journal of Sensors, 2021, Article 6657709. <https://onlinelibrary.wiley.com/doi/10.1155/2021/6657709>

Ibara, K., Hirakawa, M., & Kanetsuna, K. (2013). *Identifying Individuals' Footsteps Walking on a Floor Sensor Device*. Active Media Technology (AMT 2013), Lecture Notes in Computer Science. <https://www.researchgate.net/publication/295102931>

Jiang, M., & Yuan, J. (2025). *A device-free high-precision indoor positioning and tracking method based on GMM-WKNN algorithm*. Scientific Reports, 15, Article 17586. <https://www.nature.com/articles/s41598-025-02876-1>

Klir, S., Lerch, J., Benkner, S., & Khanh, T. Q. (2025). *Multi-Person Localization Based on a Thermopile Array Sensor with Machine Learning and a Generative Data Model*. Sensors, 25(2), 419. <https://pmc.ncbi.nlm.nih.gov/articles/PMC11769272/>

Li, X., Zhang, D., Lv, Q., Xiong, J., Li, S., Zhang, Y., & Mei, H. (2017). *IndoTrack: Device-Free Indoor Human Tracking with Commodity Wi-Fi*. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 1(3), 1-22.

Miller, R. B. (1968). *Response time in man-computer conversational transactions*. Proceedings of the AFIPS Fall Joint Computer Conference, 33, 267-277.

Naser, A., Lotfi, A., & Zhong, J. (2021). *Towards human distance estimation using a thermal sensor array*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8204928/>

Originally logged under the surname "Aziz" during source gathering; the actual first author is Naser, corrected after verification.

Ngamakeur, K., Yongchareon, S., Yu, J., & Ur Rehman, S. (2020). *A Survey on Device-free Indoor*

- Localization and Tracking in the Multi-resident Environment*. ACM Computing Surveys, 53(4). <https://dl.acm.org/doi/fullHtml/10.1145/3396302>
- Nielsen, J. (1993). *Usability Engineering*. Academic Press. Response-time limits (0.1 s, 1.0 s, 10 s) also summarised in "Response Times: The 3 Important Limits," Nielsen Norman Group. <https://www.nngroup.com/articles/response-times-3-important-limits/>
- Njanda, A., Gbadouissa, J., Rădoi, E., Abba Ari, A. A., Youssef, R., & Halidou, A. (2024). *People counting using IR-UWB radar sensors and machine learning techniques*. ScienceDirect. <https://www.sciencedirect.com/science/article/pii/S2772941924000243>
- Ozturk, M. Z., Wu, C., Wang, B., & Liu, K. J. R. (2021). *RadioMic: Sound Sensing via mmWave Signals*. <https://arxiv.org/pdf/2108.03164>
- Pegoraro, J., & Rossi, M. (2022). *Human Tracking with mmWave Radars: a Deep Learning Approach with Uncertainty Estimation*. <https://arxiv.org/pdf/2205.03274>
- Rahman, M. M., Yataka, R., Kato, S., Wang, P. P., Li, P., Cardace, A., & Boufounos, P. (2024). *MMVR: Millimeter-wave Multi-View Radar Dataset and Benchmark for Indoor Perception*. <https://arxiv.org/pdf/2406.10708>
- Ramezani Akhmareh, A., Lazarescu, M. T., Bin Tariq, O., & Lavagno, L. (2016). *A Tagless Indoor Localization System Based on Capacitive Sensing Technology*. Sensors, 16(9), 1448. <https://www.mdpi.com/1424-8220/16/9/1448>
- Segers, L., Van Bavegem, D., De Winne, S., Braeken, A., Touhafi, A., & Steenhaut, K. (2015). *An Ultrasonic Multiple-Access Ranging Core Based on Frequency Shift Keying Towards Indoor Localization*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4570340/>
- Shen, L.-H., Hsiao, A.-H., Chu, F.-Y., & Feng, K.-T. (2023). *Time-Selective RNN for Device-Free Multi-Room Human Presence Detection Using WiFi CSI*. <https://arxiv.org/pdf/2304.13107>
- Shetty, A. D., Disha, B., Shubha, B., & Suryanarayana, K. (2017). *Detection and tracking of a human using the infrared thermopile array sensor — Grid-EYE*. Proceedings of the International Conference on Intelligent Computing, Instrumentation and Control Technologies (ICICT), 1490-1495.
- Shin, Y. H., & Seo, J. (2016). *Towards Contactless Silent Speech Recognition Based on Detection of Active and Visible Articulators Using IR-UWB Radar*. Sensors, 16(11), 1812. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5134471/>
- Shubha, B., & Shastrimath, V. V. D. (2022). *Real-Time Occupancy Detection System Using Low-Resolution Thermopile Array Sensor for Indoor Environment*. IEEE Access, 10, 130981-130995. <https://www.researchgate.net/publication/366350707>
- Su, Y., Hou, H. C., Lan, H., & Ma, C. Z.-H. (2024). *A High-Fidelity mmWave Radar Dataset for Privacy-Sensitive Human Pose Estimation (the mmFree-Pose dataset)*. Bioengineering. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12383697/>

- Trofimova, A. A., Masciadri, A., Veronese, F., & Salice, F. (2017). *Indoor Human Detection Based on Thermal Array Sensor Data and Adaptive Background Estimation*. Journal of Computer and Communications, 5(4), 16-28. <https://www.researchgate.net/publication/315301224>
- Wang, G., Zou, Y., Zhou, Z., Wu, K., & Ni, L. M. (2014). *We Can Hear You with Wi-Fi! (WiHear)*. Proceedings of the 20th Annual International Conference on Mobile Computing and Networking (MobiCom).
- Wang, M., Su, D., Shi, L., Liu, Y., & Valls Miro, J. (2017). *Real-time 3D Human Tracking for Mobile Robots with Multisensors*. <https://arxiv.org/pdf/1703.04877>
- Xu, C., Li, Z., Zhang, H., Rathore, A. S., Li, H., Song, C., Wang, K., & Xu, W. (2019). *WaveEar: Exploring a mmWave-based Noise-resistant Speech Sensing for Voice-User Interface*. Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys), 14-26.
- Zheng, S., Li, J., Wang, G., Ni, M., Palit, A., Montana, G., & Guan, Y. (2025). *Person Parametric Physics-informed Representation for mmWave-based Human Pose Estimation*. <https://arxiv.org/pdf/2512.23054>
- Zhu, B., He, Z., Xiong, W., Ding, G., Huang, T., & Xiang, W. (2024). *ProbRadarM3F: mmWave Radar based Human Skeletal Pose Estimation with Probability Map Guided Multi-Format Feature Fusion*. <https://arxiv.org/html/2405.05164v2>